Master CNC Codes Reference Guide

Comprehensive Table of G-Codes, M-Codes, and Machine Parameters

G - Codes (Preparatory Functions)

#	Code	Function
1	G00	Rapid Linear Movement
2	G01	Feed Linear Movement
3	G02	Clock Wise Movement
4	G03	Counter Clockwise Movement
5	G04	Dwell / Stopping Intervals
6	G05	
7	G06	
8	G07	Imaginary Axis Designation
9	G08	
10	G09	
11	G10	Data Setting
12	G11	Data setting mode cancel
13	G12	
14	G13	
15	G14	
16	G15	

M - Codes (Miscellaneous Functions)

#	Code	Function
1	M00	Program stop
2	M01	Optional stop
3	M02	End of program
4	M03	Spindle on Clockwise
5	M04	Spindle on Counter-clockwise
6	M05	Spindle Stop
7	M06	Tool Change
8	M07	
9	M08	Coolant On
10	M09	Coolant off
11	M10	Clamp
12	M11	Unclamp
13	M12	
14	M13	
15	M14	
16	M15	

Other Machine Codes

#	Code	Function
1	X	X- Axis (Linear)
2	Y	Y - Axis (Linear)
3	Z	Z - Axis (Linear)
4	C	C - Axis (Degree)
5	I	I- Axis
6	J	J- Axis
7	K	K- Axis
8	Q	Distance for Step Movement
9	D	Daimeter
10	S	Speed
11	F	Feed
12	P	Dwell Time
13	O	Program Naming
14	N	Program Line
15	T	Tool Number
16	H	Offset Number

G - Codes (Preparatory Functions)		
#	Code	Function
17	G16	
18	G17	Establishes axis movement in the X & Y axis planes
19	G18	Establishes axis movement in the X & Z axis planes
20	G19	Establishes axis movement in the X & Y axis planes
21	G20	Values are in Inches
22	G21	Values are in Millimeters
23	G22	Return to Refrence Position
24	G23	
25	G24	
26	G25	
27	G26	
28	G27	
29	G28	
30	G29	
31	G30	Second reference position
32	G31	
33	G32	
34	G33	Thread cutting

M - Codes (Miscellaneous Functions)		
#	Code	Function
17	M16	
18	M17	
19	M18	
20	M19	
21	M20	
22	M21	
23	M22	
24	M23	
25	M24	
26	M25	
27	M26	
28	M27	
29	M28	
30	M29	
31	M30	End of program and rewind to beginning of program
32	M31	
33	M32	
34	M33	

Other Machine Codes		
#	Code	Function
17	L	Number of Repitition
18	K	Number of Repeats
19	R	Retract Plane
20	W	Home of Machine
21	U	Home of Machine

G - Codes (Preparatory Functions)		
#	Code	Function
35	G34	
36	G35	
37	G36	
38	G37	
39	G38	
40	G39	
41	G40	Cancel cutter compensation
42	G41	Cutter compensation left
43	G42	Cutter compensation Right
44	G43	Tool length compensation positive
45	G44	Tool length compensation Cancel
46	G45	Tool length compensation Negative
47	G46	
48	G47	
49	G48	
50	G49	
51	G50	
52	G51	

M - Codes (Miscellaneous Functions)		
#	Code	Function
35	M34	
36	M35	
37	M36	
38	M37	
39	M38	
40	M39	
41	M40	
42	M41	
43	M42	
44	M43	
45	M44	
46	M45	
47	M46	
48	M47	
49	M48	
50	M49	
51	M50	
52	M51	

Other Machine Codes		
#	Code	Function

G - Codes (Preparatory Functions)		
#	Code	Function
53	G52	
54	G53	Machine Coordinate Move
55	G54	Use workshift offset #1
56	G55	Use workshift offset #2
57	G56	Use workshift offset #3
58	G57	Use workshift offset #4
59	G58	Use workshift offset #5
60	G59	Use workshift offset #6
61	G60	Single direction positioning
62	G61	
63	G62	
64	G63	
65	G64	
66	G65	Macro call
67	G66	Macro modal call
68	G67	Macro modal call Cancel
69	G68	
70	G69	
71	G70	
72	G71	

M - Codes (Miscellaneous Functions)		
#	Code	Function
53	M52	
54	M53	
55	M54	
56	M55	
57	M56	
58	M57	
59	M58	
60	M59	
61	M60	
62	M61	
63	M62	
64	M63	
65	M64	
66	M65	
67	M66	
68	M67	
69	M68	
70	M69	
71	M70	
72	M71	

Other Machine Codes		
#	Code	Function

G - Codes (Preparatory Functions)		
#	Code	Function
73	G72	
74	G73	Peck drilling cycle
75	G74	
76	G75	
77	G76	Fine boring cycle
78	G77	
79	G78	
80	G79	
81	G80	Canned cycle cancel
82	G81	Drilling cycle or spot boring cycle
83	G82	Drilling cycle or counter boring cycle
84	G83	Peck drilling cycle
85	G84	Tapping cycle
86	G85	Boring cycle
87	G86	Boring cycle
88	G87	Back boring cycle
89	G88	Boring cycle
90	G89	Boring cycle
91	G90	Absolute measurements

M - Codes (Miscellaneous Functions)		
#	Code	Function
73	M72	
74	M73	
75	M74	
76	M75	
77	M76	
78	M77	
79	M78	
80	M79	
81	M80	
82	M81	
83	M82	
84	M83	
85	M84	
86	M85	
87	M86	
88	M87	ATC Leveling
89	M88	
90	M89	
91	M90	

Other Machine Codes		
#	Code	Function

G - Codes (Preparatory Functions)		
#	Code	Function
92	G91	Incrmental measurements
93	G92	
94	G93	
95	G94	Feed per minute
96	G95	Feed per revolution of the spindle
97	G96	Constant surface speed control
98	G97	Constant surface speed control Cancel
99	G98	Return to initial point in canned cycle
100	G99	Return to R point is canned cycle

M - Codes (Miscellaneous Functions)		
#	Code	Function
92	M91	
93	M92	
94	M93	
95	M94	
96	M95	
97	M96	
98	M97	
99	M98	Call SubProgram
100	M99	End Subprogram

Other Machine Codes		
#	Code	Function